

Errata

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This document contains corrections to the submitted version of the Master's thesis *Calibration of UAV thermal imagery and its influence on evapotranspiration estimation in a montane peatland: a case study of Úpské rašeliniště*.

Corrections to the text

Section 4.2.1, Flight campaigns: flight overview table. The *TCPs* column of the flight overview table lists the number of temperature calibration plates per flight incorrectly.

- Submitted: 22 June = 6; 7 August = 6; 13 August = 7; 14 August = 6.
- Corrected: 22 June = 5; 7 August = 6; 13 August = 6; 14 August = 6.

Five plates were deployed in June and all six in August. Of these, two plates are always placed in water and provide the cold reference; this is stated in the text but was not reflected in the table.

Section 5.3, Energy balance and evapotranspiration: table caption (Bowen ratio and evaporative fraction).

- Submitted: “ β and ET are dimensionless.”
- Corrected: “ β and EF are dimensionless.”

Section 5.3, Energy balance and evapotranspiration: class shift values.

- Submitted: “whereas bare peat shifted the least on each flight (by 6 %, 32 % and 25 %).”
- Corrected: “whereas bare peat shifted the least on each flight (by 6 %, 33 % and 25 %).”

Citations and cross-references

Section 2.1.1, Thermal sensor: unresolved citation.

- Submitted: “The resulting error can be significant, with differences of more than 2 °C reported between the edges and the centre of an image (Kelly et al., 2019, ?).”
- Corrected: “... between the edges and the centre of an image (Kelly et al., 2019) (Figure 1).”

Section 2.1.2, Mosaicking of thermal imagery: self-reference.

- Submitted: “... which complicates both the mosaicking of individual frames into orthomosaics and the geometric calibration of the camera (Section 2.1.2).” (*the sentence referenced its own section*)
- Corrected: “... and the geometric calibration of the camera (Casas-Mulet et al., 2020).”

Section 2.2, Theoretical framework: duplicated citation.

- Submitted: “Atmospheric correction for UAVs is also recommended and applied in Wan et al. (2024); Aubrecht et al. (2016); Duan et al. (2020); Sobrino et al. (2008); Wan et al. (2024); Kelly et al. (2019).” (*Wan et al., 2024 appears twice.*)
- Corrected: “Atmospheric correction for UAVs is also recommended and applied in Aubrecht et al. (2016); Duan et al. (2020); Sobrino et al. (2008); Wan et al. (2024) and Kelly et al. (2019).”

Section 4.1.3, Reference data: figure reference.

- Submitted: “The skin temperature of the painted aluminium plate is measured from the bottom by a PT100 sensor in the middle of the plate Figure 7.”
- Corrected: “... by a PT100 sensor in the middle of the plate (Figure 7).”

Section 4.2.1, Flight campaigns: unresolved table reference.

- Submitted: “All of the flights were conducted in stable conditions, as evidenced by ??.”
- Corrected: “... as evidenced by Table 3.”

Section 4.2.3, Empirical line calibration: unresolved table reference.

- Submitted: “Since only a limited number of reference targets were available per flight (??), ...”
- Corrected: “Since only a limited number of reference targets were available per flight (Table 3), ...”

Section 4.2.5, Orthomosaicking: figure reference.

- Submitted: “... was delineated by a normalised blue-red difference, thresholded and hole-filled Figure 12.”
- Corrected: “... thresholded and hole-filled (Figure 12).”

Figures not referenced in the text

The following figures appear in the thesis but are never cited in the body text. In each case the correction is the addition of an in-text reference to the figure at the relevant point in the given section.

Section 4.1.1, UAV data. “DJI Matrice 350 RTK with a mounted MicaSense Altum-PT sensor.” - not referenced in the text.

Section 4.1.2, Weather station. “The weather station installed at the study site.” - not referenced in the text.

Section 6.4, Reference measurements. “Global (shortwave) radiation at Luční bouda around each flight window.” - not referenced in the text.

Spelling and grammar

Section 2.1.4, Ground references.

As printed	Corrected
Regardless their form, effective references should have a high and known emissivity	Regardless of their form, effective references should have a high and known emissivity

Section 2.2, Theoretical framework.

As printed	Corrected
wavelenth	wavelength
radttiation	radiation
iamages	images
emmissivity	emissivity
trade off	trade-off
Thus, the spectral emissivity equals to spectral absorptivity	Thus, the spectral emissivity equals the spectral absorptivity
for the water and vegetation, which often has high emissivity	for the water and vegetation, which often have high emissivity

Section 4.1.3, Reference data.

As printed	Corrected
alumunium	aluminium (<i>all occurrences</i>)

Section 4.2.2, Data preprocessing.

As printed	Corrected
the median of water radiometric temperature is extracted from a radius of 5 meters where x and y are the the indices of the template pixels	... from a radius of 5 metres where x and y are the indices of the template pixels

Section 5.1, Thermal calibration.

As printed	Corrected
On each flight, the LOO-RMSE was higher the LOO-MAE. ... resulted to a MAE of 1.99 °C and a RMSE of 2.57 °C	On each flight, the LOO-RMSE was higher than the LOO-MAE. ... resulted in a MAE of 1.99 °C and a RMSE of 2.57 °C

Section 5.3, Energy balance and evapotranspiration.

As printed	Corrected
The difference between the modeled and measured net radiation footprint was less than 10 % in 3 of 4 flights.	The difference between the modelled and measured net radiation footprint was less than 10 % in 3 of 4 flights.

Section 6.1, Sensor calibration and LST calculation.

As printed	Corrected
... comparable to studies using the ELC approach/ Gerchow et al. (2025) reach an MAE of 1.5 °C and RMSE of 2 °C with a their developed repeated calibration scheme	... comparable to studies using the ELC approach. Gerchow et al. (2025) reach an MAE of 1.5 °C and RMSE of 2 °C with their developed repeated calibration scheme

Section 7, Conclusion.

As printed	Corrected
apllication themal estimation of ET through a OSEB model among the first works to present an open-source thermal UAV pipelines applied in a montane peatland	application thermal estimation of ET through an OSEB model among the first works to present an open-source thermal UAV pipeline applied in a montane peatland